

ST117 Individual DRAFT Written Report - Phase 2 Task 1 (Woodland Vegetation – Tree Measurements)

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a)

include an output of three rows

```
wl_wide <- wl %>%
  filter(FIELDNAME %in% c("DIAMETER", "HEIGHT", "DISTANCE", "NUM_STEMS")) %>%
  pivot_wider(
    id_cols = c(SITECODE, SYEAR, TREEID, STEMID, TREE_SPEC),
    names_from = FIELDNAME,
    values_from = VALUE,
    values_fn = mean
  ) %>%
  select(SITECODE, SYEAR, TREE_SPEC, DIAMETER, HEIGHT, DISTANCE, NUM_STEMS)

head(wl_wide, 3)
```

```
## # A tibble: 3 x 7
##   SITECODE SYEAR TREE_SPEC DIAMETER HEIGHT DISTANCE NUM_STEMS
##   <chr>    <int>   <int>   <dbl>  <dbl>   <dbl>   <dbl>
## 1 T05      1994     237     19    25     NA       1
## 2 T05      1994     237     16    22     NA       1
## 3 T05      1994     237     23    25     NA       1
```

b)

Limit: 50 words + 1 table.

```
model1 <- lm(HEIGHT ~ DIAMETER, data = wl_wide)

kable(summary(model1)$coefficients, digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.397	0.301	27.866	0
DIAMETER	0.109	0.009	12.018	0

Diameter is chosen as the predictor and height as the response, since taller trees generally have wider trunks. The model is statistically significant with $p < 2.2e-16$ and a slope of 0.109. However, R-squared is low (0.15), showing that diameter does not explain much of the variation in height.

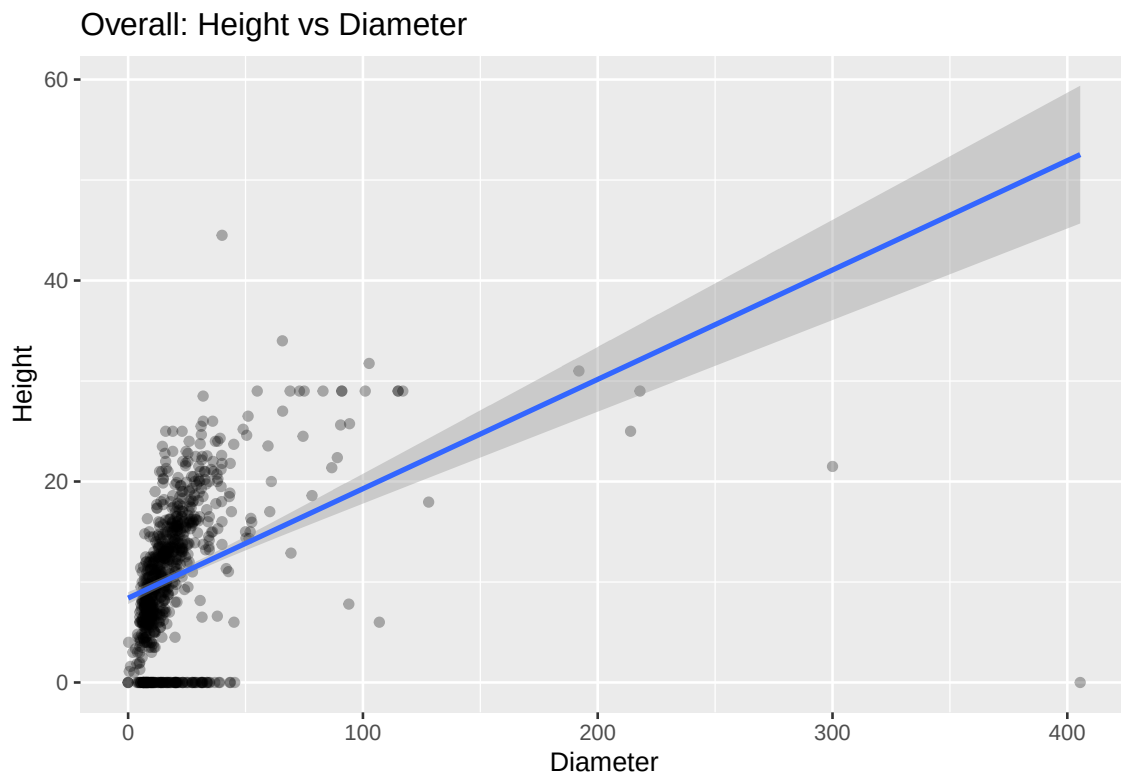
c)

Limit: 400 words + 10 figures + 5 tables

```
T05 <- wl_wide %>% filter(SITECODE == "T05")
T06 <- wl_wide %>% filter(SITECODE == "T06")
T08 <- wl_wide %>% filter(SITECODE == "T08")
T09 <- wl_wide %>% filter(SITECODE == "T09")
```

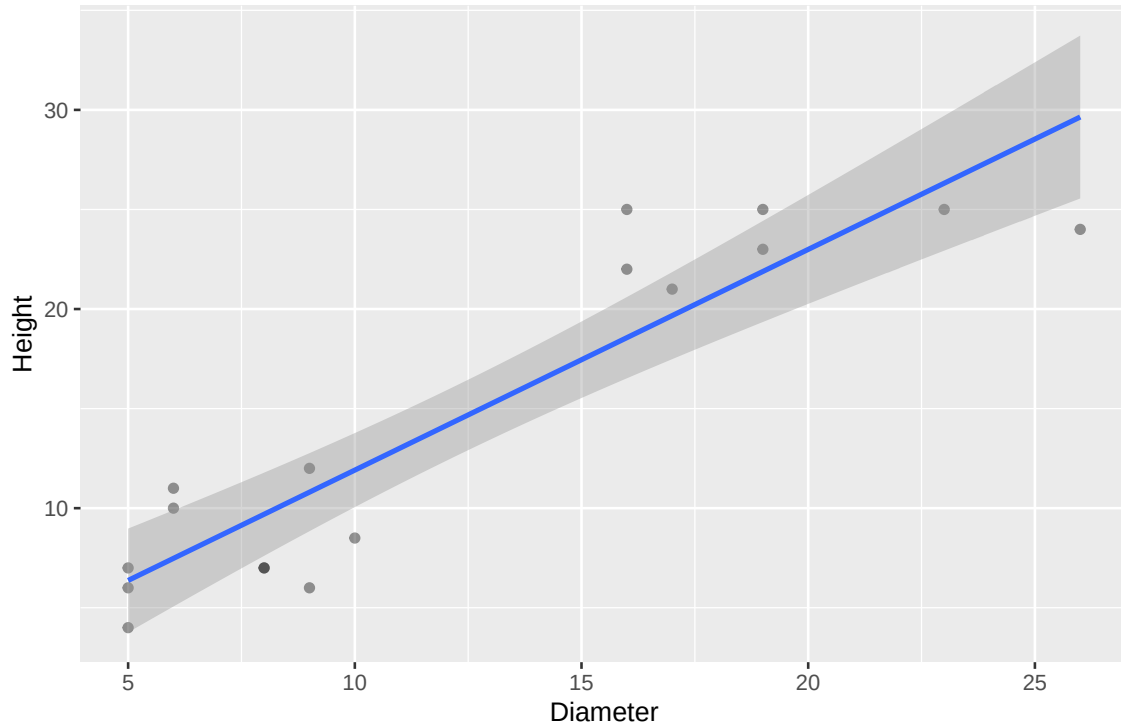
```
model_T05 <- lm(HEIGHT ~ DIAMETER, data = T05)
model_T06 <- lm(HEIGHT ~ DIAMETER, data = T06)
model_T08 <- lm(HEIGHT ~ DIAMETER, data = T08)
model_T09 <- lm(HEIGHT ~ DIAMETER, data = T09)
```

```
ggplot(wl_wide, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Overall: Height vs Diameter", x = "Diameter", y = "Height")
```



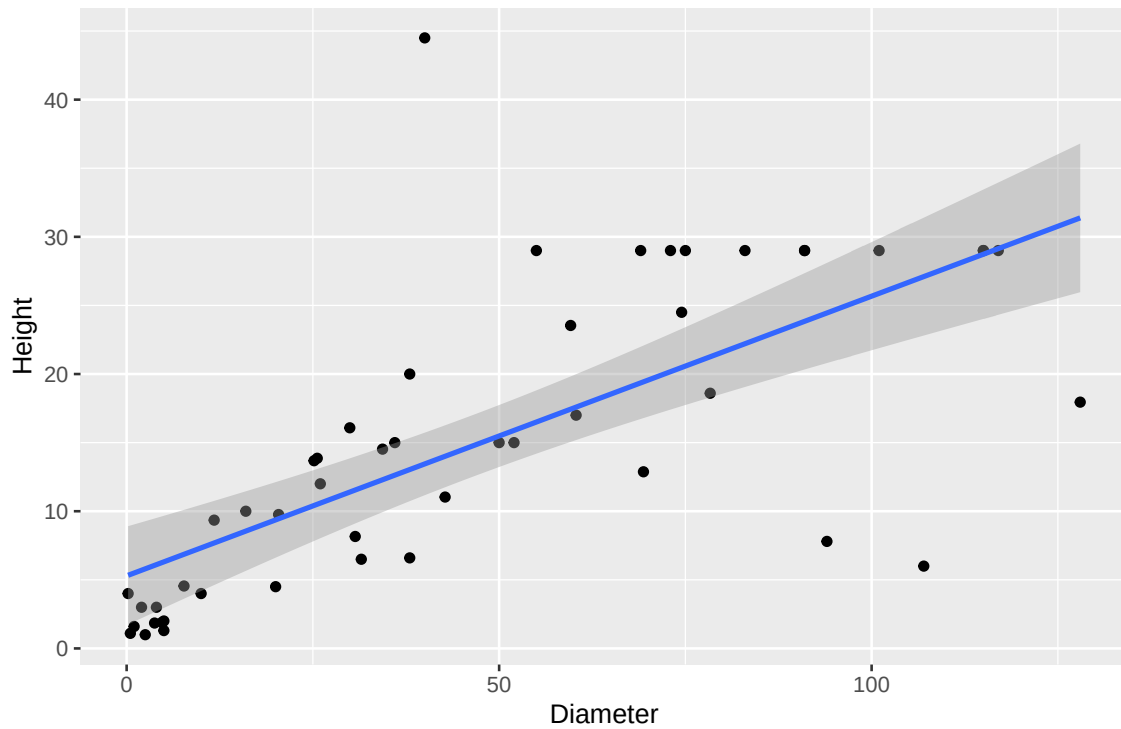
```
ggplot(T05, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(title = "T05: Height vs Diameter", x = "Diameter", y = "Height")
```

T05: Height vs Diameter

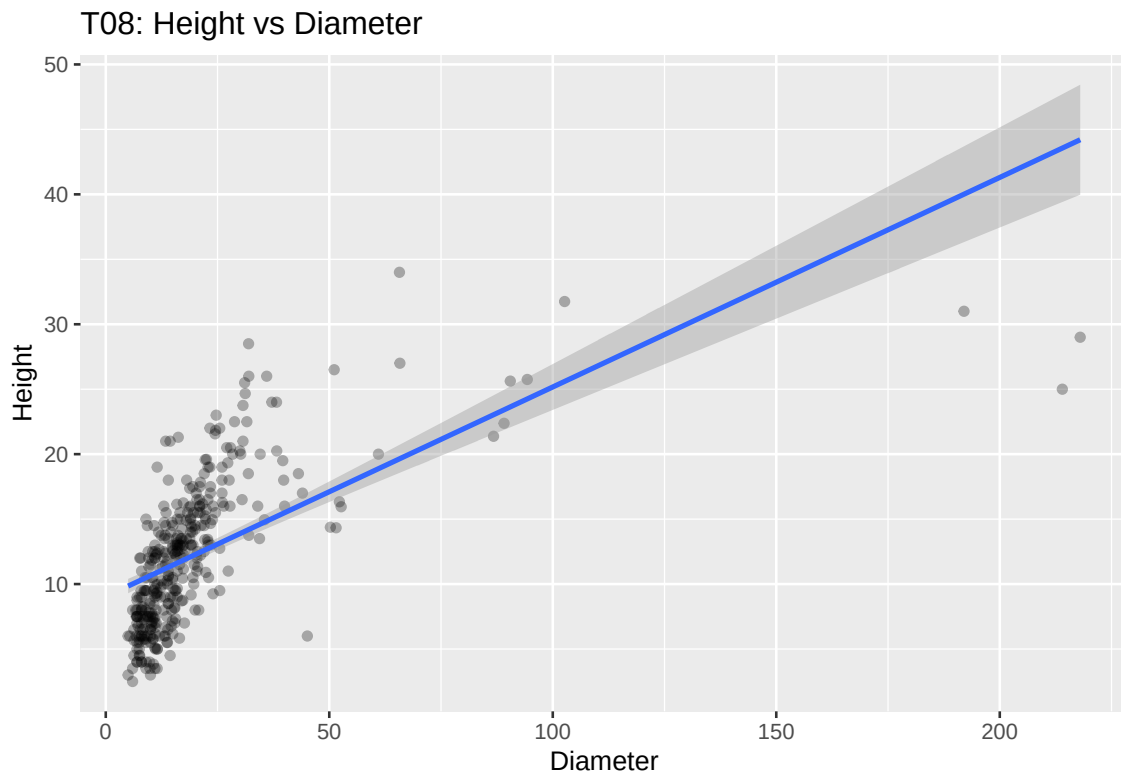


```
ggplot(T06, aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  labs(title = "T06: Height vs Diameter", x = "Diameter", y = "Height")
```

T06: Height vs Diameter

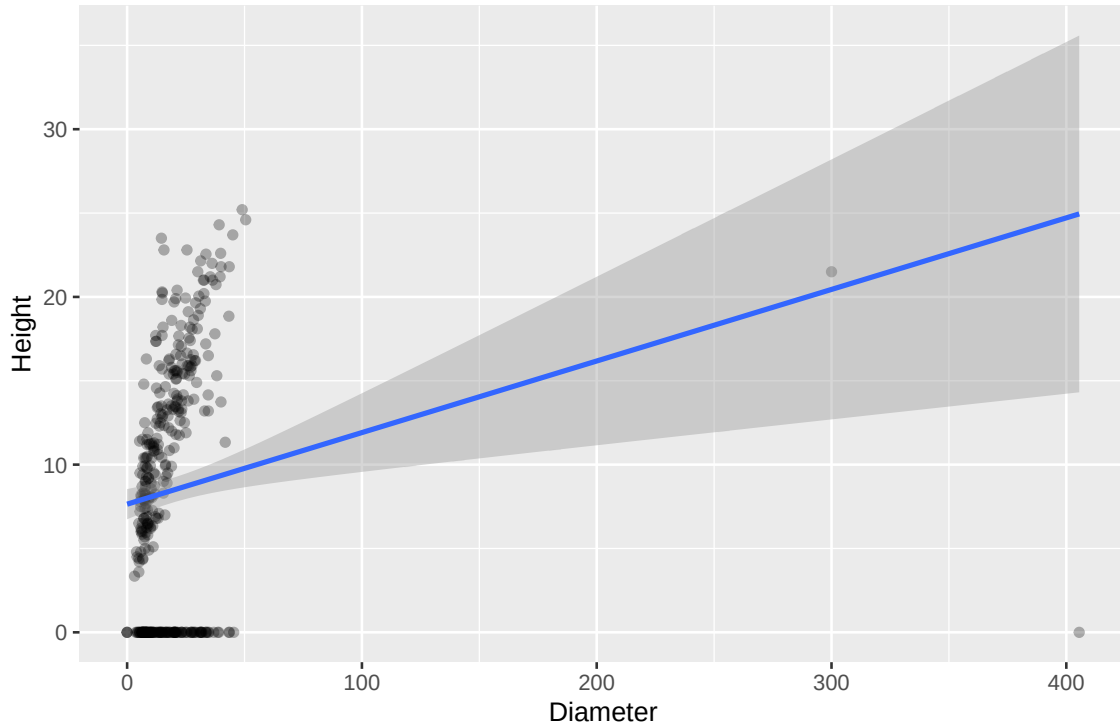


```
ggplot(T08, aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm") +  
  labs(title = "T08: Height vs Diameter", x = "Diameter", y = "Height")
```

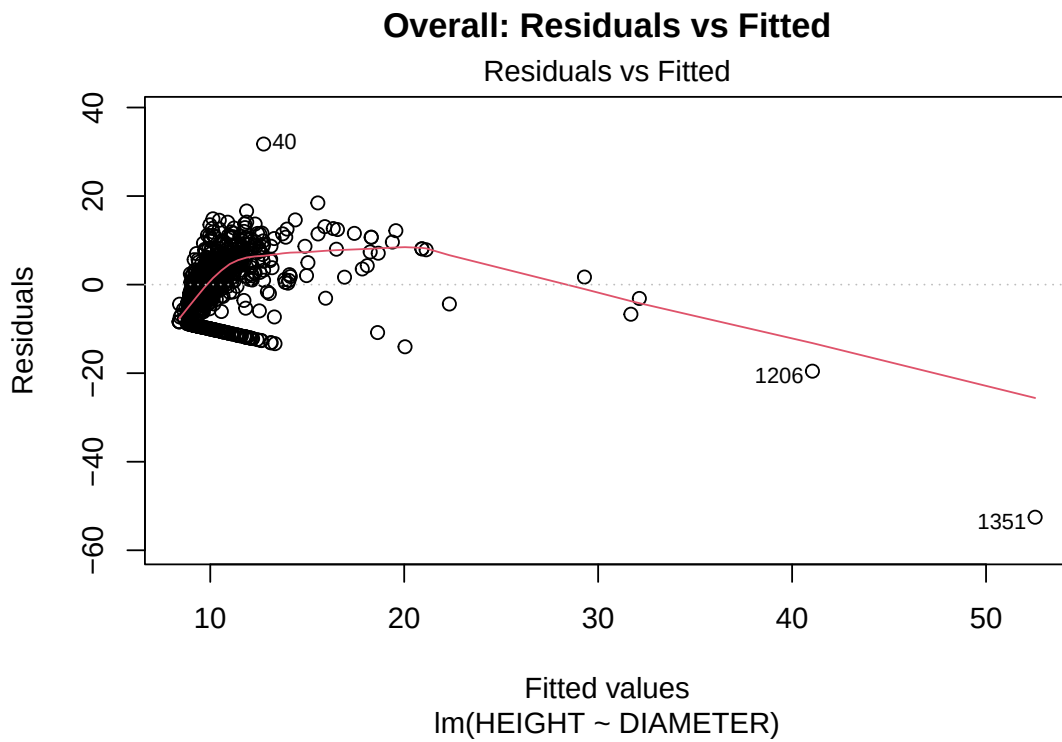


```
ggplot(T09, aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm") +  
  labs(title = "T09: Height vs Diameter", x = "Diameter", y = "Height")
```

T09: Height vs Diameter

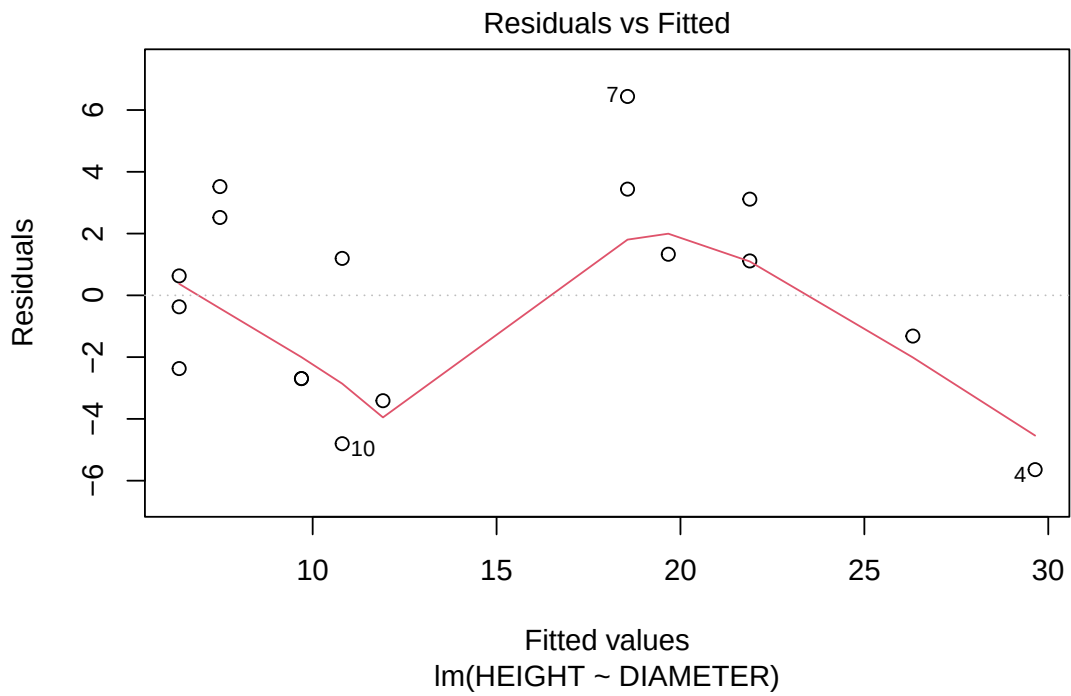


```
plot(model1, which = 1, main = "Overall: Residuals vs Fitted")
```



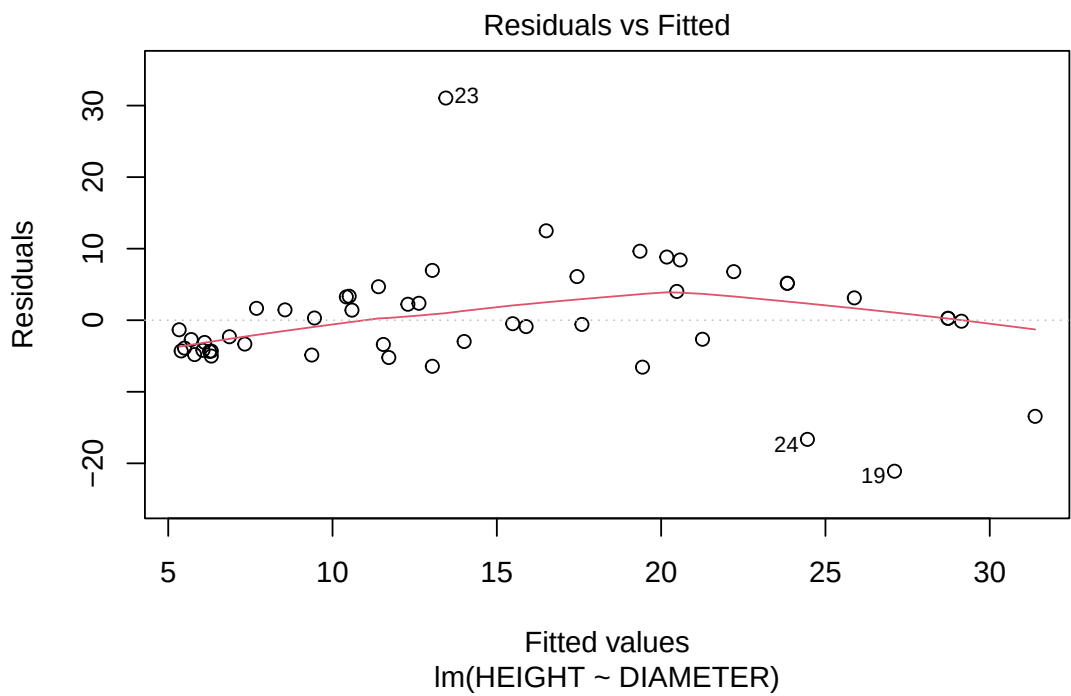
```
plot(model_T05, which = 1, main = "T05: Residuals vs Fitted")
```

T05: Residuals vs Fitted



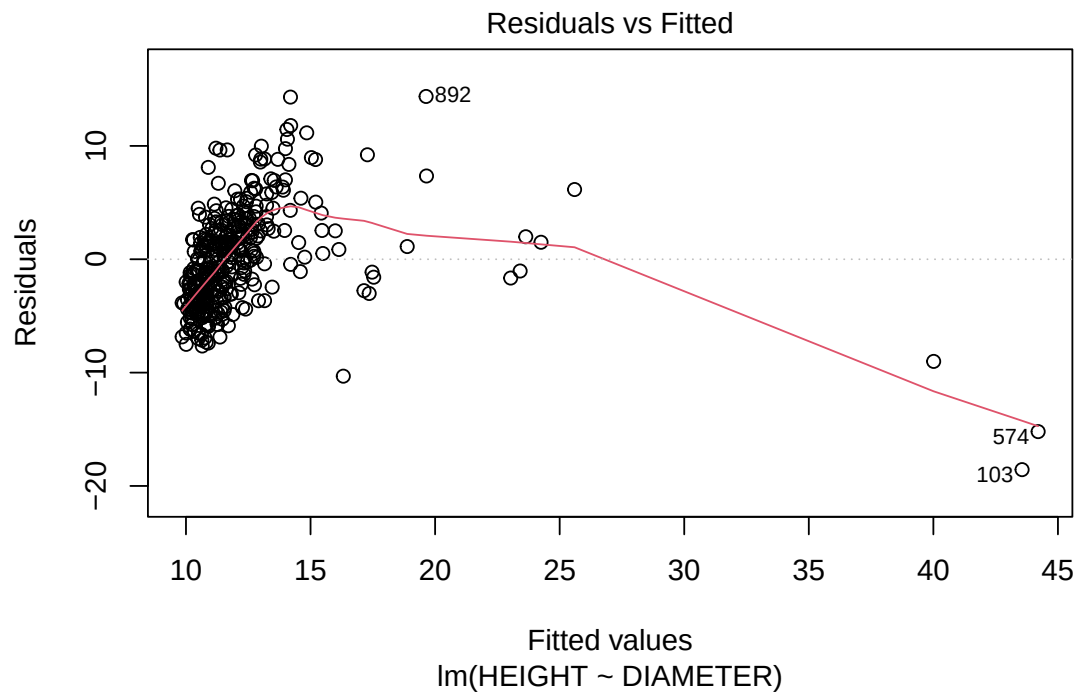
```
plot(model_T06, which = 1, main = "T06: Residuals vs Fitted")
```

T06: Residuals vs Fitted



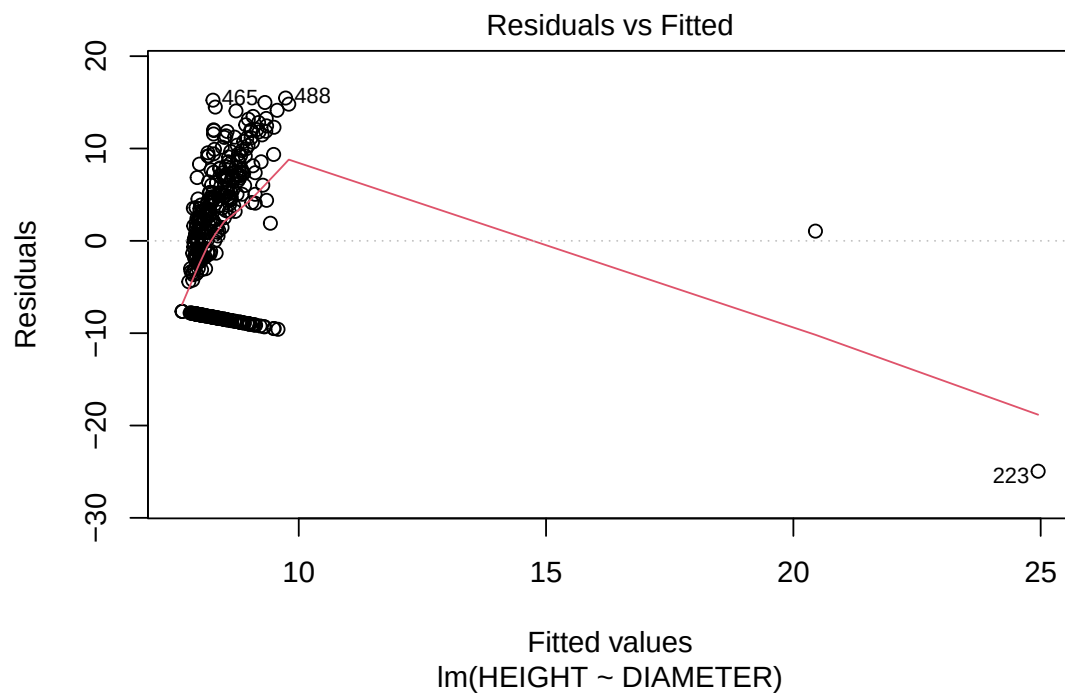
```
plot(model_T08, which = 1, main = "T08: Residuals vs Fitted")
```

T08: Residuals vs Fitted



```
plot(model_T09, which = 1, main = "T09: Residuals vs Fitted")
```

T09: Residuals vs Fitted



```
kable(summary(model_T05)$coefficients, digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.829	1.736	0.478	0.64
DIAMETER	1.108	0.125	8.848	0.00

```
kable(summary(model_T06)$coefficients, digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.292	1.783	2.967	0.005
DIAMETER	0.204	0.030	6.794	0.000

```
kable(summary(model_T08)$coefficients, digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.040	0.315	28.653	0
DIAMETER	0.161	0.011	14.944	0

```
kable(summary(model_T09)$coefficients, digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.643	0.458	16.693	0.000
DIAMETER	0.043	0.014	3.058	0.002

```
data.frame(
  Model = c("Overall", "T05", "T06", "T08", "T09"),
  R2 = c(summary(model1)$r.squared, summary(model_T05)$r.squared,
    summary(model_T06)$r.squared, summary(model_T08)$r.squared,
    summary(model_T09)$r.squared),
  Slope = c(coef(model1)[2], coef(model_T05)[2], coef(model_T06)[2],
    coef(model_T08)[2], coef(model_T09)[2])
) %>% kable(digits = 3)
```

Model	R2	Slope
Overall	0.153	0.109
T05	0.839	1.108
T06	0.495	0.204
T08	0.382	0.161
T09	0.025	0.043

There is a positive correlation between diameter and height in all models, since large diameter trees usually would be taller. But there seems to be some variance among all models in how well the relationship between diameter and height holds in different monitored sites.

T05 has the best fit, with a slope of 1.108, showing that diameter has significant effects on height. On the other hand, T06 has a moderate fit, while T08 has a low fit. Other factors may affect the relationship between diameter and height in these two areas.

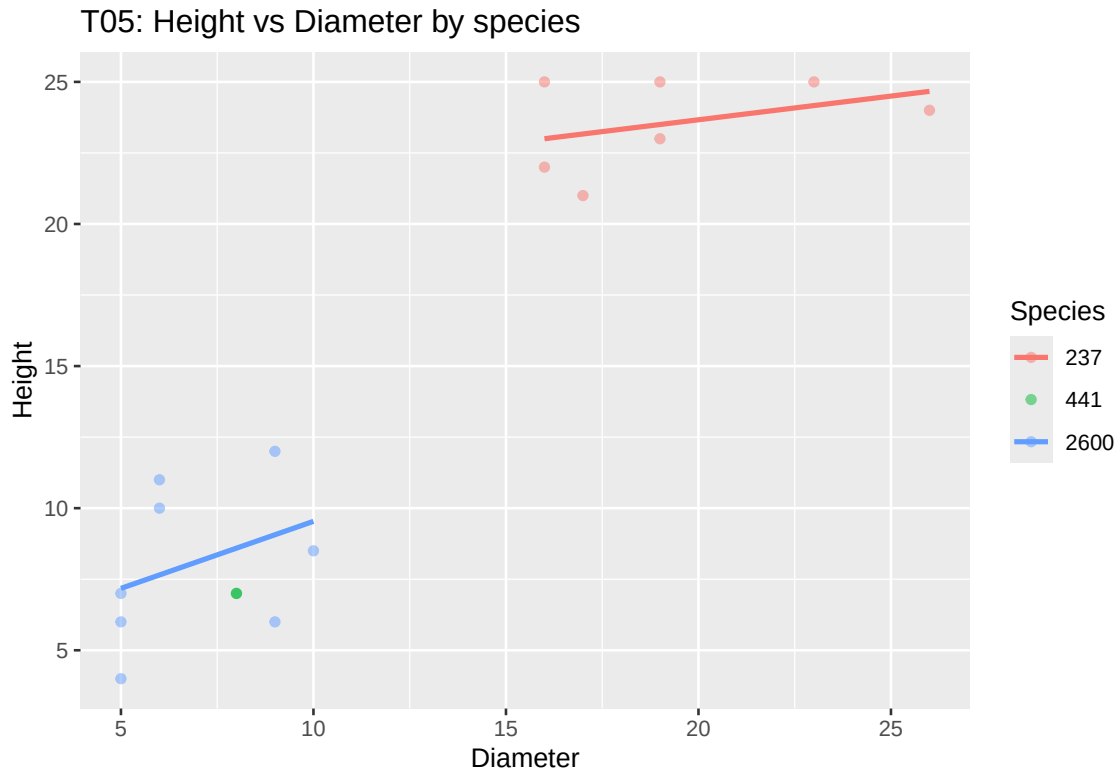
In T09, there is almost no relationship between diameter and height. This may be because the area is more diverse with different species growing differently and have varied environmental factors.

When considering the overall model, the model does not fit well relative to individual models. By pooling data, site-level variance is averaged out. The residual plots of T08 and T09 are similar to the overall model, while those of T05 and T06 are not as similar.

d)

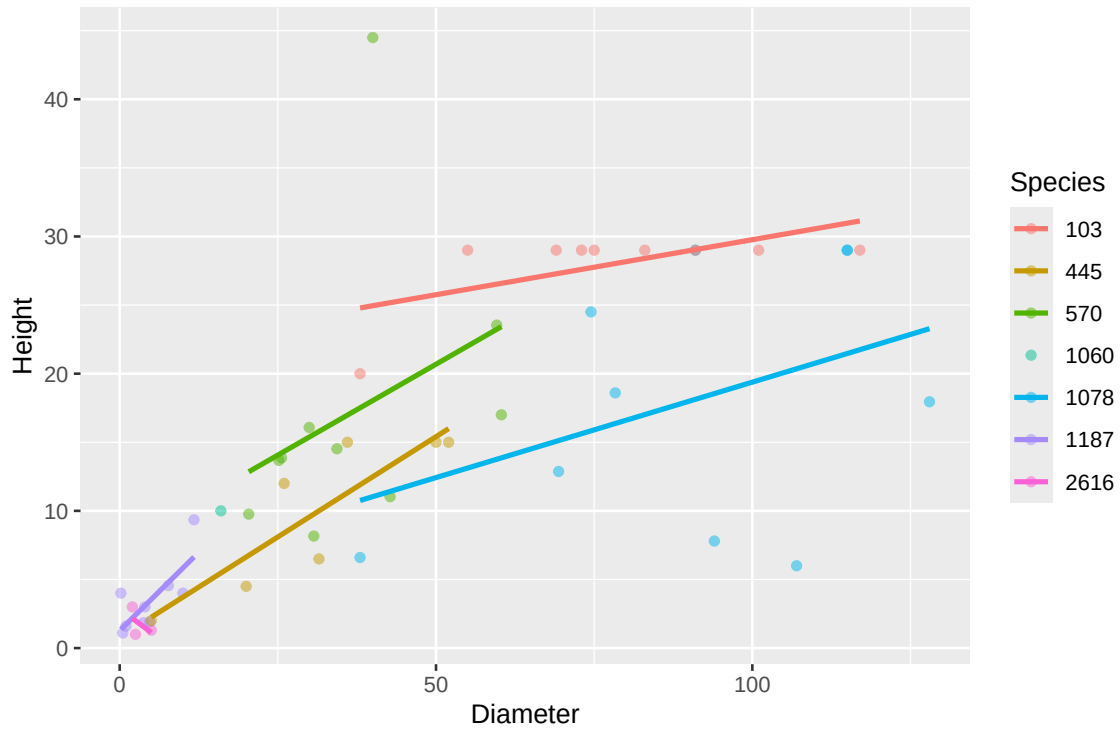
Limit: 300 words + 4 figures + 4 tables

```
ggplot(T05, aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "T05: Height vs Diameter by species", x = "Diameter", y = "Height", colour = "Species")
```



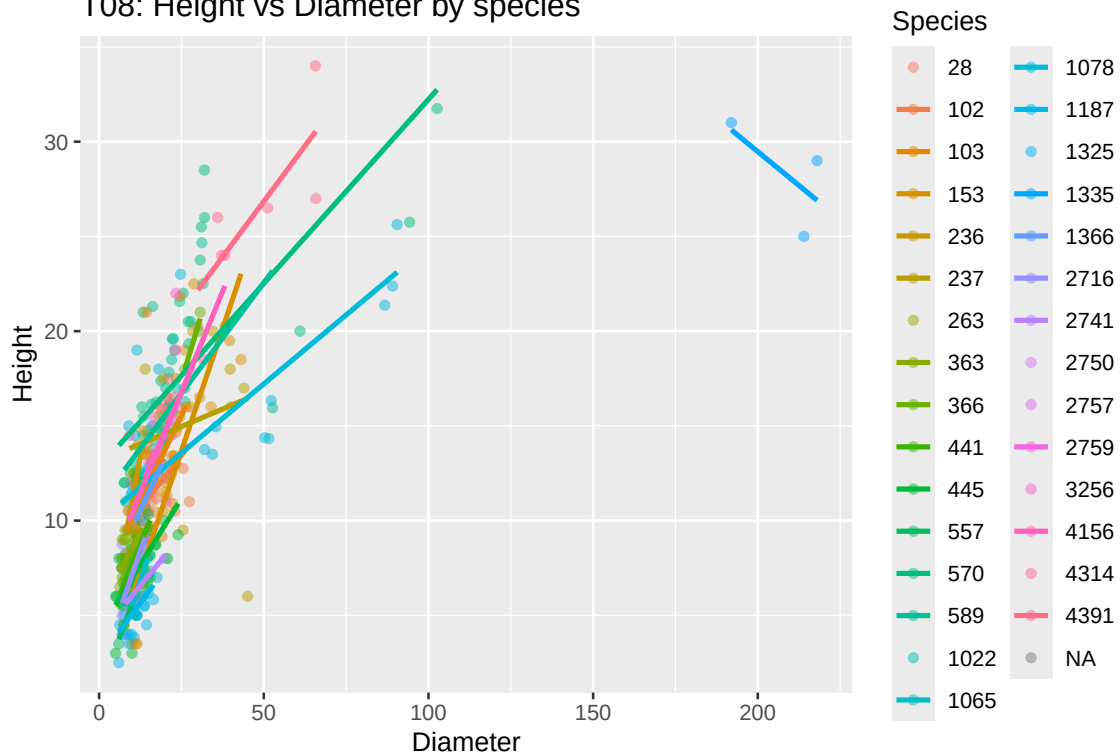
```
ggplot(T06, aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "T06: Height vs Diameter by species", x = "Diameter", y = "Height", colour = "Species")
```

T06: Height vs Diameter by species

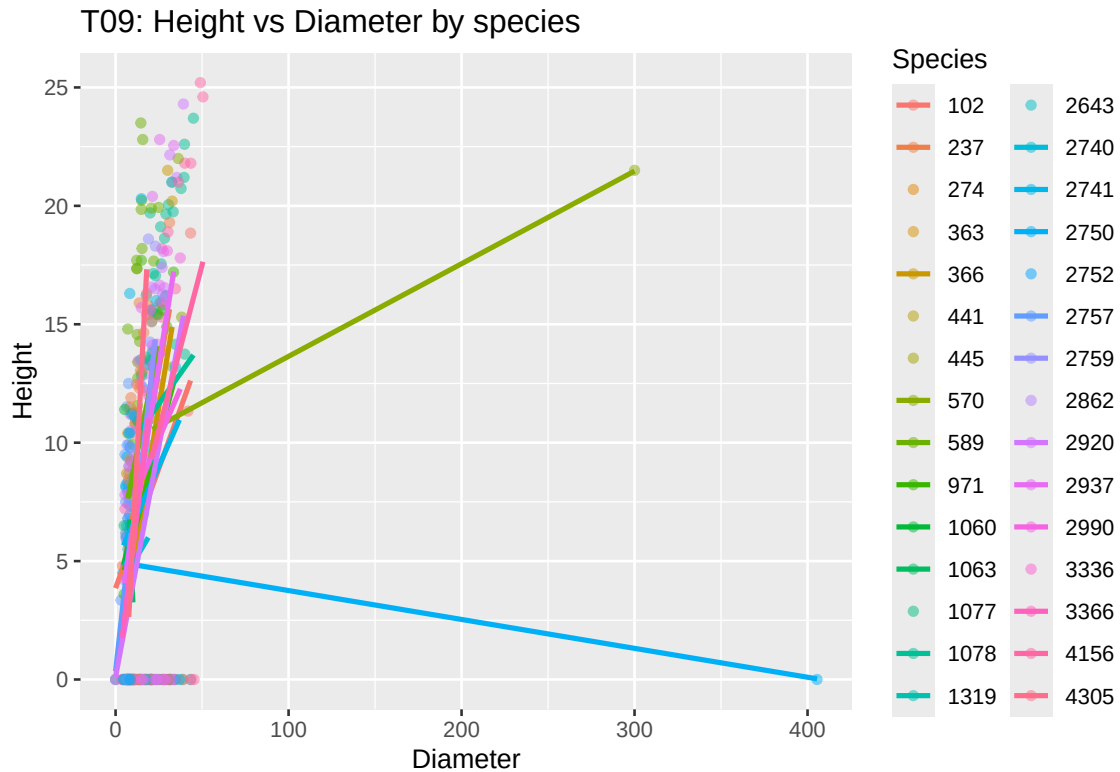


```
ggplot(T08, aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T08: Height vs Diameter by species", x = "Diameter", y = "Height", colour = "Species")
```

T08: Height vs Diameter by species



```
ggplot(T09, aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T09: Height vs Diameter by species", x = "Diameter", y = "Height", colour = "Species")
```



```
model_T05_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC), data = T05)
model_T06_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC), data = T06)
model_T08_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC), data = T08)
model_T09_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC), data = T09)

data.frame(
  Site = c("T05", "T06", "T08", "T09"),
  R2 = c(summary(model_T05)$r.squared, summary(model_T06)$r.squared,
    summary(model_T08)$r.squared, summary(model_T09)$r.squared),
  R2_species = c(summary(model_T05_sp)$r.squared, summary(model_T06_sp)$r.squared,
    summary(model_T08_sp)$r.squared, summary(model_T09_sp)$r.squared)
) %>% kable(digits = 3)
```

Site	R2	R2_species
T05	0.839	0.942
T06	0.495	0.691
T08	0.382	0.770
T09	0.025	0.130

One issue affecting all four locations is the fact that there are several types of trees with varying diameter-height slopes. A straight regression line that would be applicable for all species would not be ideal because different species have different diameters and heights.

For location T05, species 2600 shows higher growth than species 237 as indicated by a steeper slope while species 441 has only one value so it makes no sense to plot a regression line for it. Species 570

is the dominant species for T06 and determines the slope of the line. In T08, the number of species exceeds 30, thus, contributing to the high scatter of data around the line.

Incorporation of species into the model as a categorical variable will improve the fit of data on all locations. The values of R-squared increase from 0.84 to 0.94 in T05, 0.50 to 0.69 in T06, 0.38 to 0.77 in T08, and 0.025 to 0.13 in T09.